PHYS 7501:
Particles and Nuclei I

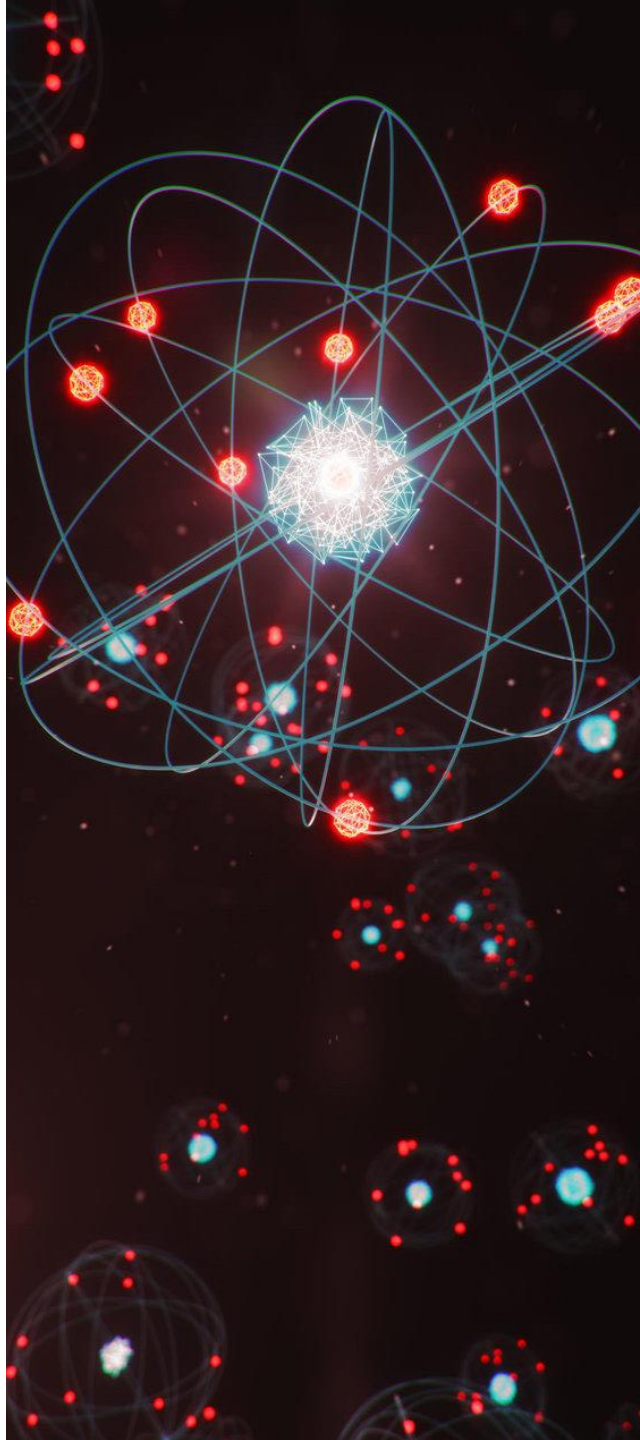
Tues./Thurs. 2 – 3:20PM
Clippinger 105

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NUCLEAR PHENOMENOLOGY

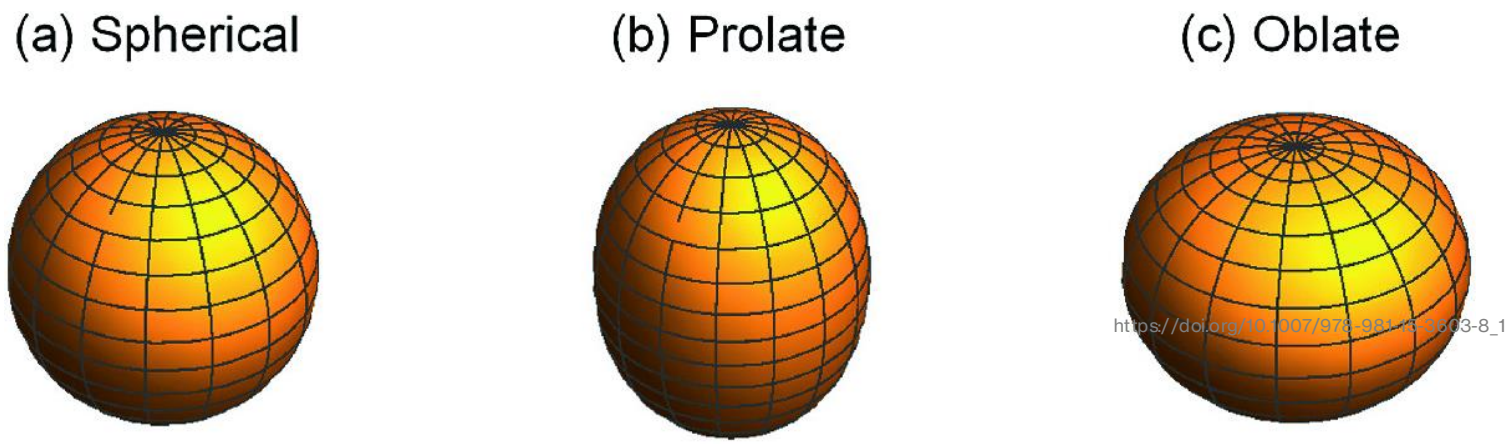
Modern Nuclear Chemistry, Chapter 2

Basic Ideas and Concepts in Nuclear Physics, Heyde, Chapter 1

Phenomenology refers to empirically motivated models!

Nuclear Shapes

- The **strong** nuclear interaction establishes the distribution and motion of nucleons in the nucleus, and we probe that distribution with the **electromagnetic** interaction.
- The shape of nuclei can be inferred from their **electromagnetic moments**



- Nuclear magnetic moments describe the distribution of electric currents in the nucleus
- Nuclear electric moments describe the distribution of electric charges in the nucleus

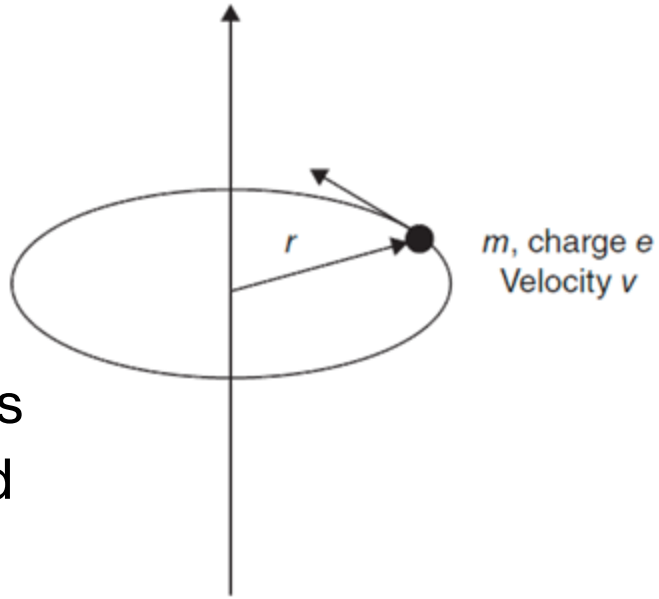
A moment is a distance multiplied by a physical quantity. For distributions, you integrate the quantity with respect to distance.

Moments

- Electric charges and currents produce electric and magnetic fields that vary with distance in a characteristic way
- Charge and current distributions have an **electromagnetic multipole moment** associated with each characteristic spatial dependence
  - $\frac{1}{r^2}$ - field arises from net charge referred to as a zeroth or monopole moment (no magnetic monopoles exist)
  - $\frac{1}{r^3}$ - field arises from the first or dipole moment
  - $\frac{1}{r^4}$ - field arises from the second or quadrupole moment
- Generally, the most important moments are the **electric quadrupole moment Q** and **magnetic dipole moment μ**

Magnetic Dipole Moment

- The magnetic moment of a nucleus is a measure of the average electric current
- To understand the magnetic moment, we can start classically
- Consider an electron moving with velocity v in a circular orbit with radius r
- The magnetic dipole moment of this moving charge is defined as the product of the area of the loop made by the electron (A) and the current (i): $\mu = iA$
- Current can be defined as the charge of the electron divided by the time to complete the loop: $i = \frac{e}{\frac{2\pi r}{v}}$



- Combining these terms:

$$|\mu| = iA = \frac{ev}{2\pi r} \times \pi r^2 = \frac{evr}{2}$$

The absolute value sign here is to indicate that the electron has a direction

- The angular momentum of an electron moving in a circle is $\ell\hbar = m_0 vr$,

$$|\mu| = \frac{e\ell\hbar}{2m_0}$$

$\frac{e}{2m_0}$ is a constant called the gyromagnetic ratio and is denoted as γ

We can remove the absolute value since we know that the projection of the angular momentum is $m_\ell\hbar$ and includes the direction of rotation

$$|\mu| = \gamma\ell\hbar$$

$$\mu = \left(\frac{e}{2m_0}\right) m_\ell\hbar = m_\ell\mu_B$$

Constants are included in one number (μ_B) called the Bohr magneton. $\mu_B = 5.78 \times \frac{10^{-5} ev}{Tesla}$

- Extending this description to protons and neutrons, the **proton** will have an intrinsic magnetic moment due to its spin and an additional part if it has orbital motion. The **neutron** will only have the intrinsic magnetic moment.
- We can define a **nuclear magneton** as

$$\mu_N = \frac{e\hbar}{2m_p} = 3.15 \times 10^{-8} ev/Tesla$$

Traditionally, we expand this definition by including a constant of proportionality called the gyromagnetic ratio or *g*-factor

$$\mu_\ell^{proton} = m_\ell \mu_N$$

For protons, $g_\ell = 1$
For neutrons, $g_\ell = 0$

$$\mu = g_\ell m_\ell \mu_N$$

- This is only for orbital motion, what about spin?

- Both protons and neutrons have intrinsic spin so we can expect additional contributions to the total magnetic moment with the form:

$$\mu_s = g_s m_s \mu_N$$

$m_s = 1/2$ for protons and neutrons

- Precise measurements of g_s yield

Proton: $g_s = 5.5856912(22)$

Neutron: $g_s = -3.8260837(18)$

- If the proton and neutron were structure-less, you'd expect values of 2 and 0.
- The total magnetic moment is then

$$\boldsymbol{\mu = (g_\ell m_\ell + g_s m_s)\mu_N}$$

$$\boldsymbol{\mu_{tot} = (\mu_\ell + \mu_s)}$$

μ_N is treated as unit and left off

Magnetic Moment of ^{15}N

- Make an estimate of the ground state magnetic moment of ^{15}N if the nuclear spin $I = \frac{1}{2}$ is due to an unpaired proton in an $\ell = 0$ orbital

$$\boldsymbol{\mu_{tot} = (\mu_\ell + \mu_s)}$$

- All nucleons are paired except one proton, so we only need to consider the proton here

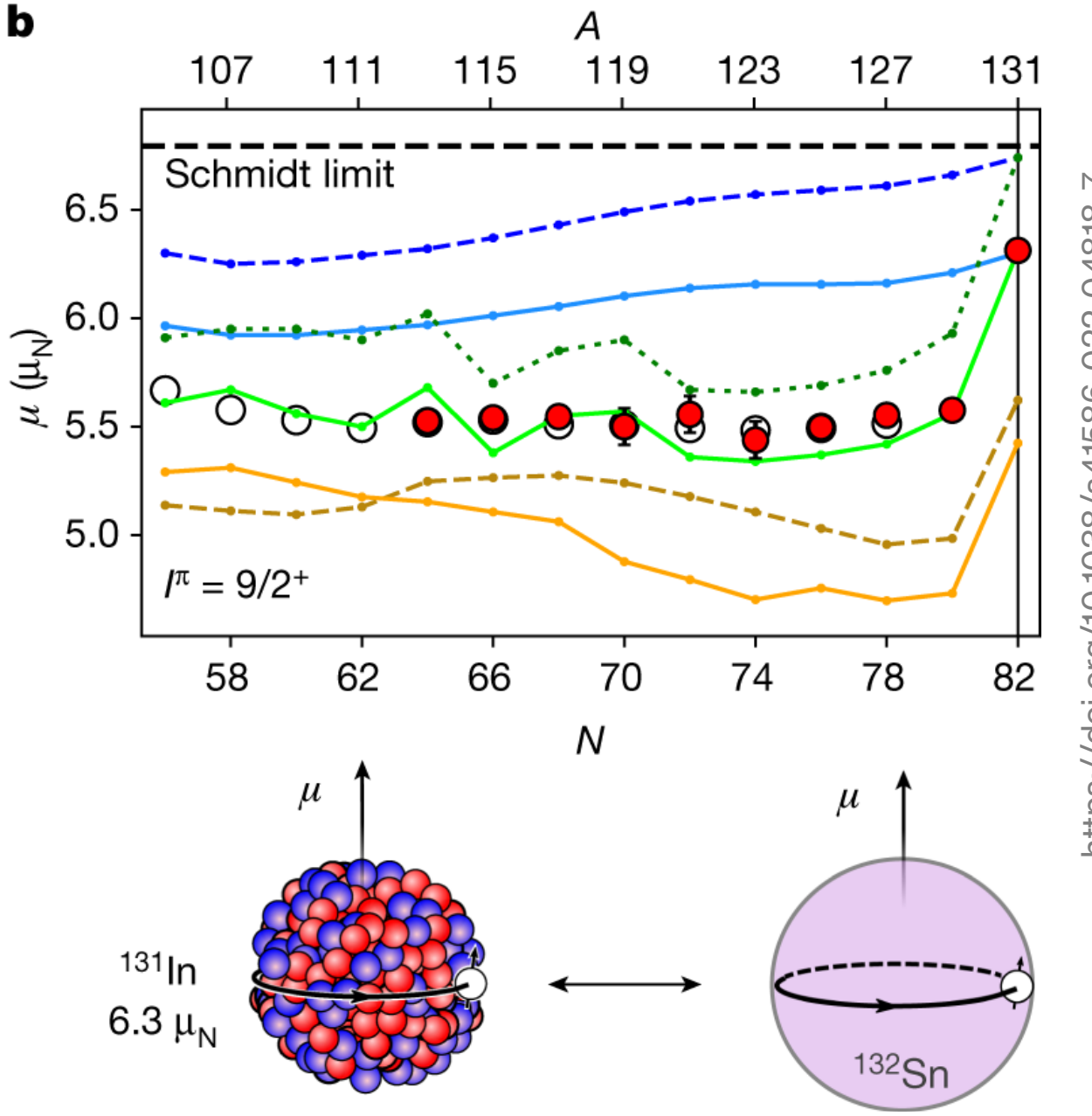
$$\boldsymbol{\mu_{tot} = (g_\ell m_\ell \mu_N + g_s m_s \mu_N)}$$

$$\boldsymbol{\mu_{tot} = \left(1 \times 0\mu_N + 5.5856912 \times {}^{1}/_{2}\,\mu_N\right)}$$

$$\boldsymbol{\mu_{tot} = 2.7928456\mu_N}$$

Magnetic Moment Trends

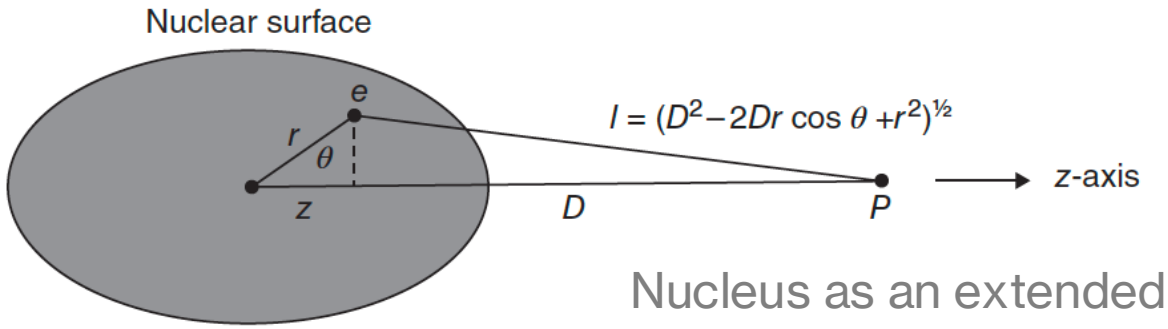
- When we discuss the Nuclear Shell Model next week, you'll learn more about these and how to calculate them



https://doi.org/10.1038/s41586-022-04818-7

Electric Quadrupole Moment

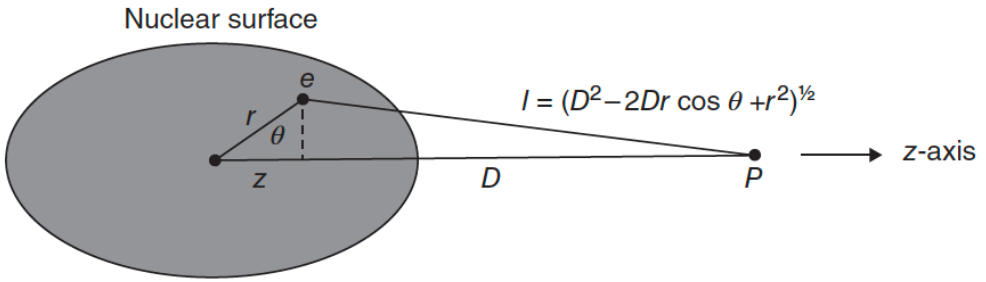
- The electric moment is a measure of the **distribution of the electric charge** in a nucleus
- So far, we've assumed nuclei were spherical with uniform electric charge distributions
- If a nucleus could be symmetrically stretched or squashed, it would yield a **quadrupole charge distribution**



Nucleus as an extended charged object

- A nonzero electric quadrupole moment of a nucleus implies a nonspherical charge distribution

- Calculate the electric potential energy at some point $\boldsymbol{P}$, which is a distance $\boldsymbol{D}$ from the center of the charged nucleus
- Evaluate the potential $\boldsymbol{d\Phi}$ at point $\boldsymbol{P}$ due to a charge at a distance $\boldsymbol{r}$ from the center of the charge object at an angle $\boldsymbol{\theta}$
- The density of charge is given by $\boldsymbol{\rho(\theta, \phi, r)}$



$\boldsymbol{\theta, \phi, r}$ are defined with respect to the nuclear centroid

$$\boldsymbol{d\Phi} = \frac{\boldsymbol{\rho(\theta, \phi, r)}\,\mathrm{d}\boldsymbol{\tau}}{\ell}$$

$$\boldsymbol{d\Phi} = \frac{\boldsymbol{\rho(\theta, \phi, r)}\,\mathrm{d}\boldsymbol{\tau}}{\sqrt{[\boldsymbol{D^2 + r^2 - 2Dr\cos\theta}]}}$$

Factor out D

$$\boldsymbol{d\Phi} = \frac{\boldsymbol{\rho(\theta, \phi, r)}\,\mathrm{d}\boldsymbol{\tau}}{\boldsymbol{D}}\left(\boldsymbol{1 - 2Dr}\cos\boldsymbol{\theta} + \frac{\boldsymbol{r^2}}{\boldsymbol{D^2}}\right)^{\boldsymbol{-1/2}}$$

Taylor expanding the quantity in parentheses yields a multipole expansion

$$\boldsymbol{d\Phi} = \frac{\boldsymbol{\rho(\theta, \phi, r)}\,\mathrm{d}\boldsymbol{\tau}}{\boldsymbol{D}}\left(\boldsymbol{1} + \frac{\boldsymbol{2}\cos\boldsymbol{\theta}}{\boldsymbol{2}}\frac{\boldsymbol{r}}{\boldsymbol{D}} + \frac{\boldsymbol{1}}{\boldsymbol{8}}\boldsymbol{(3(2}\cos\boldsymbol{\theta})^{\boldsymbol{2}}\boldsymbol{-4)}\frac{\boldsymbol{r^2}}{\boldsymbol{D^2}} + \ \ldots\right)$$

$$d\Phi = \frac{\rho(\theta,\phi,r)\,\mathrm{d}\tau}{D}\left(1 + \frac{2\cos\theta}{2}\frac{r}{D} + \frac{1}{8}\left(3(2\cos\theta)^2 - 4\right)\frac{r^2}{D^2} + \ \ldots\right)$$

$$d\Phi = \frac{\rho(\theta,\phi,r)\,\mathrm{d}\tau}{D}\left(1 + \frac{r}{D}\cos\theta + \left(\frac{3}{2}(\cos\theta)^2 - \frac{1}{2}\right)\frac{r^2}{D^2} + \ \ldots\right)$$

- Substituting in the Legendre Polynomials, $P_1(\cos\theta) = \cos\theta$ and $P_2(\cos\theta) = \frac{3}{2}(\cos\theta)^2 - \frac{1}{2}$

$$d\Phi = \frac{\rho(\theta,\phi,r)\,\mathrm{d}\tau}{D}\left(1 + \frac{r}{D}P_1(\cos\theta) + \frac{r^2}{D^2}P_2(\cos\theta) + \ \ldots\right)$$

- Integrating the expression over the entire volume of the charged object, all odd Legendre terms drop-out:

$$\Phi = \frac{1}{D}\left[\int \rho\,\mathrm{d}\tau\right] + \frac{1}{D^2}\left[\int \rho r\cos\theta\ \mathrm{d}\tau\right] + \frac{1}{D^3}\left[\int \rho r^2\left(\frac{3}{2}(\cos\theta)^2 - \frac{1}{2}\right)\mathrm{d}\tau + \cdots\right]$$

Recall that $d\tau = dr\sin\theta d\theta d\phi$

Electric dipole moment

$$\Phi = \frac{1}{D}\left[\int \rho \mathrm{d}\tau\right] + \frac{1}{D^2}\left[\int \rho \mathrm{r}\cos\theta \ \mathrm{d}\tau\right] + \frac{1}{D^3}\left[\int \rho r^2\left(\frac{3}{2}(cos\,\theta)^2 - \frac{1}{2}\right)\mathrm{d}\tau + \cdots\right]$$

Electric monopole moment = nuclear charge Z_e **Electric quadrupole moment**

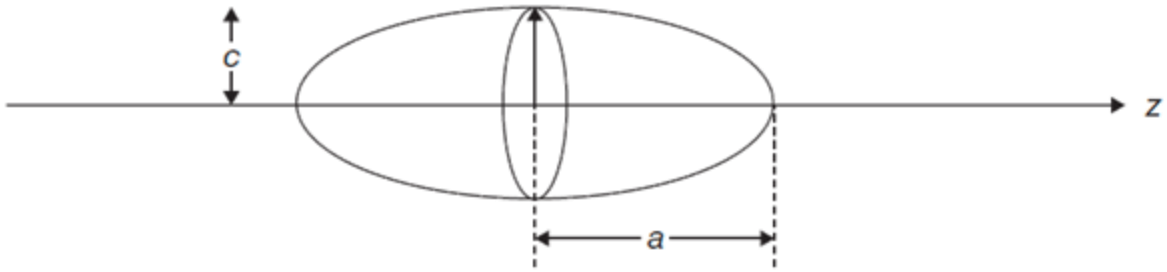
- For any well-defined quantum state, the density is an even function and since the dipole term involves the product of an even and odd, the integral is zero. There should be **NO electric dipole moment for any nuclei**.

$$Q = \iiint \rho(r) r^2 \left(\frac{3}{2}(cos\,\theta)^2 - \frac{1}{2}\right) r^2 dr \sin\theta \, d\theta d\phi$$

- For spherical nuclei, the density does not depend on θ, therefore the quadrupole moment Q is zero!

- Any non-zero Q indicates a non-spherical (deformed) nucleus
- A simple model is to assume nuclei are ellipsoids with a shape generated by rotating the ellipsoid around one of its axes
- Call the semi-minor axis c and the semi-major axis a (axis of rotation in this case)
- If a is long, we've made a prolate shape (rugby ball) and if a is the short axis, then we've made an oblate shape (flattened sphere)

$$\boldsymbol{Q = \frac{2}{5}Ze(a^2 - c^2)}$$

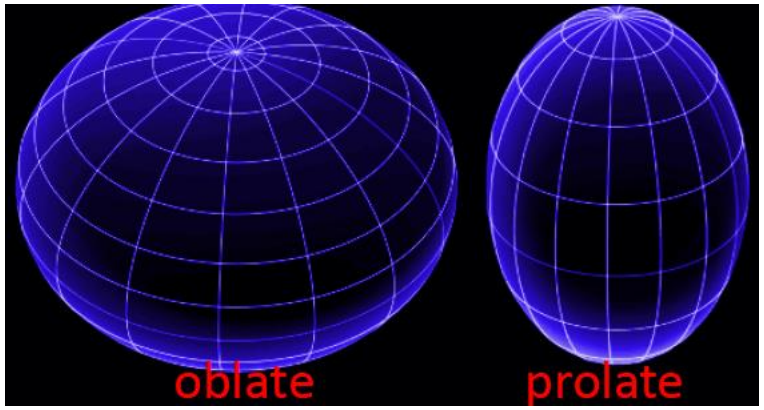


- We know that the mean radius R squared of a spheroid related by two semi-axes is:

$$\boldsymbol{R^2 = \frac{1}{2}\left(a^2 + c^2\right) = (r_0 A^{\frac{1}{3}})^2}$$

- If $a > c$, then $Q > 0$ and the nucleus is prolate
- If $a < c$, then $Q < 0$ and the nucleus is oblate

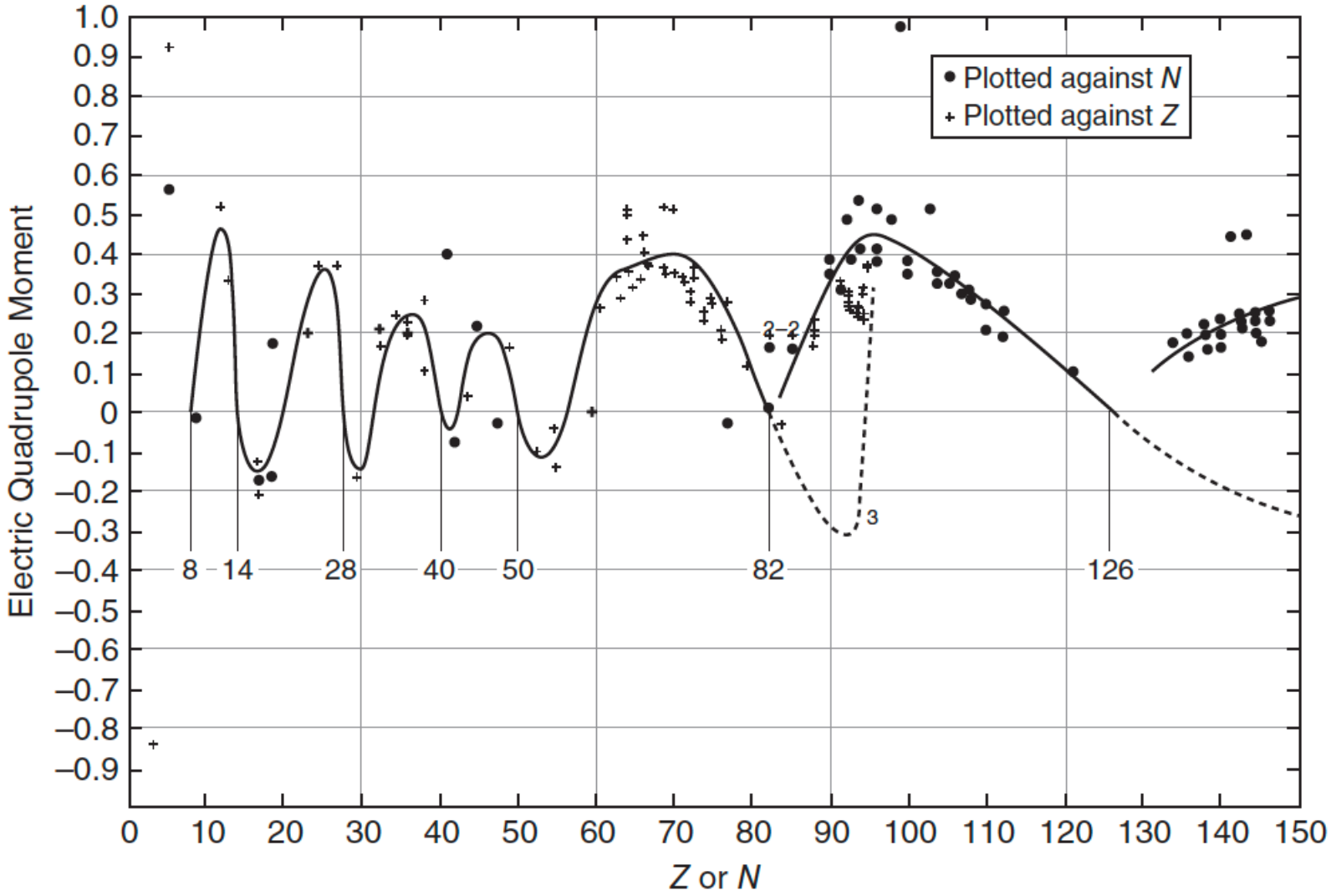
Q is generally quoted in terms of e



Electric Quadrupole Moment

- Q/e is commonly reported and has units of area
- A convenient unit for small areas is the ***barn***:

$$1\,barn = 10^{-24} cm^2$$



Higher orders of deformation also exist!



Quadrupole Moment of ^{177}Hf

- Calculate the ratio of the semi-major to semi-minor axes of the prolate nucleus ^{177}Hf given $Q = +3.0\ e \cdot barns$

$$\boldsymbol{Q = \frac{2}{5} Ze(a^2 - c^2)}$$

Hf has Z=72

$$\boldsymbol{(a^2 - c^2) = \frac{Q/e}{2Z/5} = \frac{3.0 \times 10^{-24} cm^2}{2 \times 72/5}}$$

$$\boldsymbol{= 1.04 \times 10^{-25} cm^2}$$

$$\boldsymbol{R^2 = \frac{1}{2}(a^2 + c^2) = (r_0 A^{\frac{1}{3}})^2}$$

$$\boldsymbol{(a^2 + c^2) = 2r_0^2 A^{2/3} = 2(1.2fm)^2 177^{2/3}}$$

Convert to cm

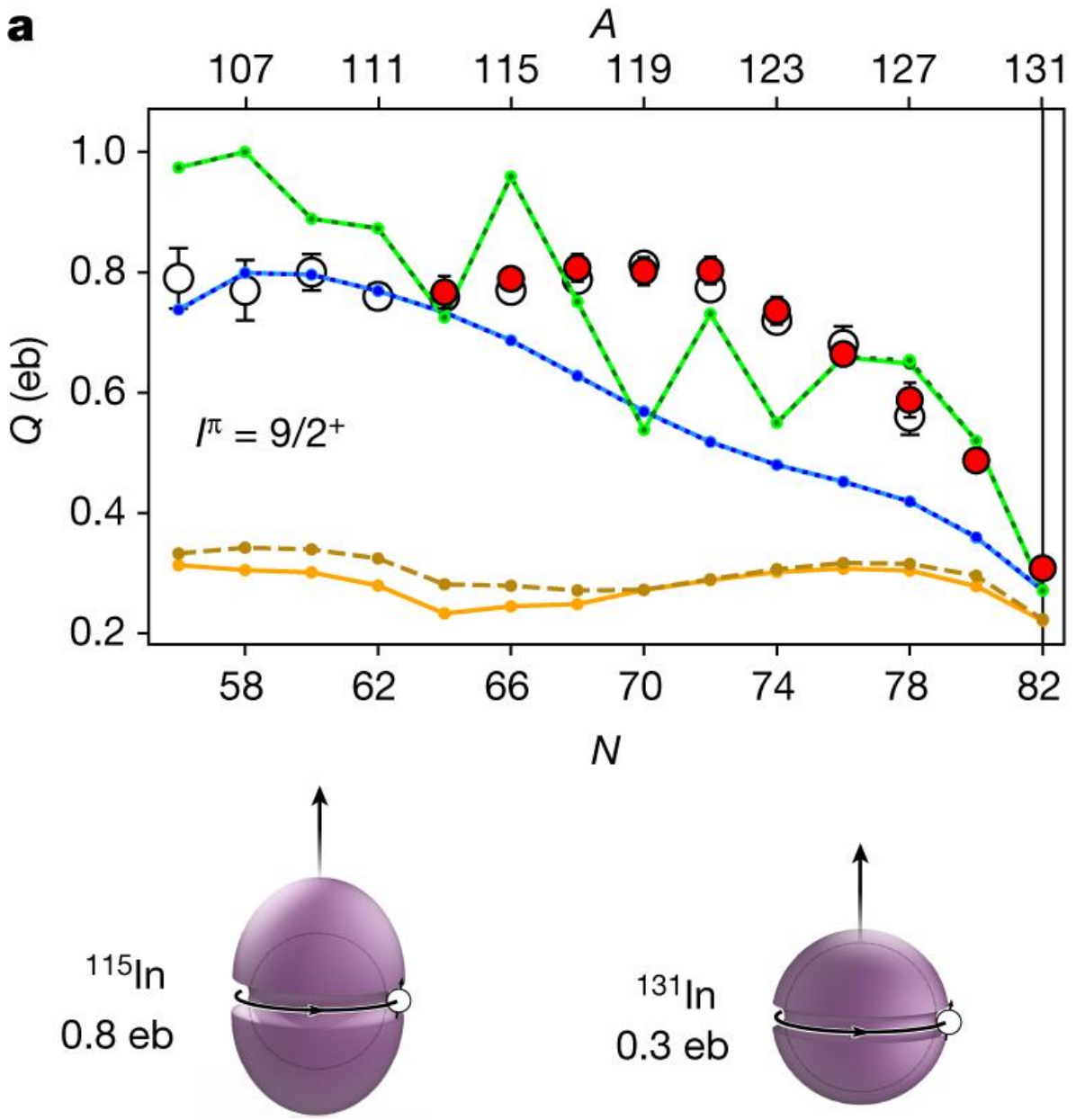
$$\boldsymbol{= 9.08 \times 10^{-25}\ cm^2}$$

$$\boldsymbol{(a^2 + c^2) + (a^2 - c^2) = 2a^2 = 1.012 \times 10^{-24} cm^2}$$
$$\boldsymbol{a = 7.1 \times 10^{-13} cm}$$
$$\boldsymbol{c = 6.3 \times 10^{-13} cm}$$

-$a > c$, then $Q > 0$ and the nucleus is prolate!!

Electric Quadrupole Moment Trends

- When we discuss the Nuclear Shell Model next week, you'll learn more about these and how to calculate them



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Wrapping up

- HW posted on GitHub in the HW folder
  - Due September 9, 11:59PM via e-mail or Edwards Mailbox (door locks at 5:30PM...)
- Office Hours on Friday at 10AM
  - Edwards 204
- **Next class**: Nuclear Structure next week!
- Questions?